1 **(a)** Positive **[B1]**

(b) Single straight ruled line passing within the bounds of the data points, with roughly equal numbers of points on either side. **[B1]**

(c) 64 (Accept values between 62 and 67 strictly based on the student's line of best fit) **[B1]**

[Total Marks: 3]

2 **(a)** Positive **[B1]**

(b) A single ruled straight line drawn crossing the data set, passing within tolerance of the mean point $(6.3, 67)$ with roughly equal points on either side. **[B1]**

(c) Accept any value strictly read from the candidate's own line of best fit at $x = 5.5$ (expected roughly 60 to 62). **[B1]**

[Total Marks: 3]

3 **(a)** Point $(2.8, 11.5)$ plotted accurately as a small cross **[B1]**

Point $(3.0, 11.0)$ plotted accurately as a small cross **[B1]**

(b) Negative **[B1]**

(c) **(i)** $\bar{x} = \dfrac{21.0}{10} = 2.1$ **[B1]**

$\bar{y} = \dfrac{145.0}{10} = 14.5$ **[B1]**

(ii) Mean point $(2.1, 14.5)$ plotted accurately **[B1]**

Single ruled straight line passing through $(2.1, 14.5)$ across the data range with roughly equal numbers of points on either side **[B1]**

(d) Answer in the range 12.4 to 13.0 (or follow-through from candidate's line at $x = 2.5$) **[B1]**

(e) 5.0 litres lies outside the range of the given data ($1.2 \leq x \leq 3.0$), so the linear trend may not continue (extrapolation is unreliable) **[B1]**

[Total Marks: 9]

4 **(a)** Plotting the remaining 6 points correctly. **[P2]**

(Allow P1 for 4 or 5 points correct).

(b) Negative **[B1]**

(c) A ruled straight line of best fit drawn by eye, with a roughly equal number of points on each side. **[B1]**

[Total Marks: 4]

5 **(a)** All 8 points plotted correctly **[B2]**

(5, 6, or 7 points plotted correctly [B1])

(b) Positive **[B1]**

(c) Ruled line passing through the mean point $(45, 45)$ and fitting the trend of the data **[B1]**

(d) Value read correctly from their line of best fit (typically $42 - 48$) **[B1]**

[Total Marks: 5]

6 (a) $y = 5.96x + 34.7$ (or $m = 5.96$, $c = 34.7$) **[B2]**

(Allow B1 for either $m = 5.96$ or $c = 34.7$ correct to 3 sf)

(b) Substitution of $x = 7$ into their equation from (a): $5.96(7) + 34.7$ **[M1]**

76.4 (or 76) **[A1ft]**

(c) The value $x = 15$ is outside the range of the given data (extrapolation). **[B1]**

[Total Marks: 5]

7 (a) Points $(7, 5)$ and $(9, 1)$ correctly plotted **[B1]**

(b) Mean age = 4.5 **[B1]**

Mean price = 11 **[B1]**

(c) Mean point $(4.5, 11)$ plotted accurately **[B1]**

Ruled line of best fit passing through $(4.5, 11)$ **[B1]**

(d) Negative (or strong negative) **[B1]**

(e) Value read correctly from their line of best fit (typically $2 - 4$) **[B1]**

[Total Marks: 7]

8 (a) All three points correctly plotted at $(31, 81)$, $(32, 85)$, and $(35, 96)$ **[B2]**

Note: Award [B1] if exactly two points are plotted correctly.

(b) Mean temperature $= \dfrac{224}{8} = 28$ **[B1]**

Mean drinks sold $= \dfrac{552}{8} = 69$ **[B1]**

(c) Mean point $(28, 69)$ clearly plotted (e.g. marked with a cross or a circled cross). **[B1]**

A ruled straight line of best fit passing directly through their mean point, with an appropriate gradient. **[B1]**

(d) Accept any value strictly read from the candidate's own line of best fit at $x = 29$ (expected roughly 72 to 75). **[B1]**

[Total Marks: 7]

9 (a) All three points $(6, 81)$, $(7, 88)$, and $(8, 100)$ plotted correctly **[B2]**

(Allow B1 if only 1 or 2 points are plotted correctly)

(b) $\bar{x} = \dfrac{35}{7} = 5$ **[B1]**

$\bar{y} = \dfrac{490}{7} = 70$ **[B1]**

(c) Mean point $(5, 70)$ correctly plotted (allow a distinct mark e.g. a circle around the cross) **[B1]**

Single straight ruled line passing exactly through their mean point, following the trend of the data **[B1]**

(d) 75 (Accept values between 73 and 77 strictly based on the student's line of best fit) **[B1]**

[Total Marks: 7]

10 (a) (i) Gradient $a = 1.0857\ldots$ rounded to 1.09 **[B1]**

y-intercept $b = 23.1428\ldots$ rounded to 23.1, giving $1.09x + 23.1$ **[B1]**

(ii) $\bar{x} = 27.5$ **[B1]**

$\bar{y} = 53$ **[B1]**

(b) (i) $y = 1.0857(32) + 23.1429$ (or $1.09(32) + 23.1$) **[M1]**

57.9 (or 58) **[A1]**

(ii) 90 minutes is outside the range of the given data ($10 \leq x \leq 45$), so it involves unreliable extrapolation **[B1]**

The model gives $y \approx 121$ marks, which is impossible if the test is out of 100 (linear growth cannot continue indefinitely) **[B1]**

(c) $\bar{x}$ decreases because $25 < 27.5$ **[B1]**

$\bar{y}$ decreases because $51 < 53$ **[B1]**

[Total Marks: 10]

11 (a) $m = -1500$ **[B1]**

$c = 14000$ **[B1]**

Note: Award [B2] for the fully correct equation $y = -1500x + 14000$.

(b) It represents the amount the car's value decreases (depreciates) per year. (Accept references to losing \$1500 each year). **[B1]**

(c) $y = -1500(4.5) + 14000$ **[M1]**

7250 **[A1]**

(d) 15 years is far outside the given data range (extrapolation) OR the equation would predict a negative value (-8500), which is impossible. **[B1]**

[Total Marks: 6]

12 (a) $\bar{x} = \dfrac{2100}{12} = 175$ **[B1]**

$\bar{y} = \dfrac{924}{12} = 77$ **[B1]**

(b) Substitution of $(\bar{x}, \bar{y})$ into the equation: $77 = 0.85(175) - c$ **[M1]**

$c = 148.75 - 77 = 71.75$ **[A1]**

(c) Predicted mass $= 0.85(180) - 71.75 = 81.25$ **[M1]**

New mean $= \dfrac{924 + 81.25}{13} = 77.3$ (or 77.32...) **[A1]**

[Total Marks: 6]

13 (a) (i) $m = -5.82$ (accept -5.81 to -5.82) **[B1]**

$c = 27.0$ (accept 27.0 to 27.1) **[B1]**

(ii) The decrease in fuel consumption (in km/l) for every 1 litre increase in engine size. **[B1]**

(b) Substitution of $x = 1.6$ into their equation: $-5.816(1.6) + 27.0$ **[M1]**

17.7 (accept 17.6 to 17.8 depending on rounding of m and c) **[A1]**

(c) Because 5.0 is outside the range of the given data (extrapolation). **[B1]**

[Total Marks: 6]

14 (a) Mean temperature = 21.25 **[B1]**

Mean ice creams sold = 80.25 **[B1]**

(b) $m = 4.28$ **[B1]**

$c = -10.6$ **[B1]**

(c) $y = 4.28 \times 24 - 10.6$ **[M1]**

92 (or 92.12) **[A1]**

(d) Unreliable because 42°C is outside the range of the given data (extrapolation) **[B1]**

[Total Marks: 7]

15 (a) $p = 8(11) - (4 + 7 + 8 + 12 + 14 + 15 + 18) = 10$ **[B1]**

$q = 8(45) - (26 + 34 + 38 + 43 + 47 + 53 + 62) = 57$ **[B1]**

(b) (i) $a = 2.62$ **[B1]**

$b = 16.18$, giving $2.62x + 16.18$ **[B1]**

(ii) For every additional \$1000 spent on advertising, monthly sales increase by \$2620 on average **[B1]**

(c) Predicted sales $= 2.62(14) + 16.18 = 52.86$ thousand dollars **[M1]**

Difference $= 53 - 52.86 = 0.14$ thousand dollars **[M1]**

140 **[A1]**

(d) (i) $68 = 2.62x + 16.18$ giving $x = \dfrac{51.82}{2.62} = 19.7786\ldots$ **[M1]**

19.8 **[A1]**

(ii) Extrapolation: $x = 19.8$ lies outside the given range of advertising spend ($4 \leq x \leq 18$) **[B1]**

[Total Marks: 11]